Moshe Mash, Ph.D.

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Academic Positions

Postdoctoral Researcher | Carnegie Mellon University, School of Computer Science, Robotics Institute | 2019–2022

- **Hosts:** Prof. Reid Simmons and Dr. Stephanie Rosenthal

My research studied the people who build AI systems: how machine learning practitioners and data scientists work as they develop models, and where they get stuck. I developed DSWorkFlow, a framework that captures data scientists' workflows from Jupyter notebooks, reconstructs the order in which code was executed, and combines it with qualitative data, and used it in studies of data scientists building machine learning models (CHI 2021). Building on that data, I developed a model that predicts stuckness from real-time indicators such as code artifacts, and an algorithm that times an intervention as close as possible to the onset of stuckness (VL/HCC 2022).

I mentored computer science students. The research was funded by J.P. Morgan, which renewed the award for a second consecutive year based on the project's results.

Education

Ph.D., Software and Information Systems Engineering | Ben-Gurion University of the Negev | 2014–2018

- **Dissertation:** Reaching Fair Agreements in Group Settings
- **Advisor:** Prof. Ya'akov (Kobi) Gal

Dissertation research on how people and computer agents reach fair agreements in group settings. One line of work addressed the classic rent division problem: assigning rooms to housemates and splitting the rent fairly. We developed a polynomial-time algorithmic framework for computing envy-free solutions that optimize a chosen fairness criterion, and proved that the maximin solution, which maximizes the minimum utility subject to envy-freeness, is also equitable, minimizing the disparity between players. We established this theoretically, empirically on real data from the Spliddit website, and in a user study with Spliddit users, and the solution has been deployed on Spliddit since 2015 (**Best Paper Award, ACM EC 2016**; IJCAI 2017; JACM 2017; **CACM Research Highlight 2018**). A second line studied coalition

formation in weighted voting games, combining cooperative game theory with machine learning to predict how people form coalitions, and designing an agent that negotiates with people and outperformed human players (IJCAI 2017; ACM TIST 2020). Mentored master's students and advised final-year projects throughout the Ph.D.

M.Sc., Computer Science (AI specialization) | Bar-Ilan University | 2011–2013

- **Thesis:** Joint Exploration with Self-Interested Agents
- **Advisor:** Prof. David Sarne

Thesis research on joint exploration by self-interested agents: settings in which several agents search for opportunities whose values are uncertain and costly to reveal, and share what they find. Using equilibrium analysis, the work showed that mechanisms intended to enhance cooperation, such as longer search horizons or more cooperating agents, can degrade individual and overall utility when agents are self-interested, with implications for the design of multi-agent markets and systems (AIJ 2014; IAT 2012).

B.Sc., Software Engineering | Shenkar College of Engineering and Design | 2007–2011

Awards & Recognition

- **Best Paper Award** – ACM Conference on Economics and Computation (2016)
- **Communications of the ACM Research Highlight** – Which Is the Fairest (Rent Division) of Them All? (2018)
- **Excellence in Teaching Award** – Ben-Gurion University (2016, 2017)
- **Ph.D. Scholarship for Academic Excellence** – Ben-Gurion University (2014)

Professional Experience

Principal AI Researcher & Founder | MashInnovateAI | 2024–present

- Independent AI research & development consultancy: helps companies identify where AI can solve their core challenges, then designs and delivers the solution end to end, from proof of concept to production.
- Clients include technology companies looking to integrate AI into their products and workflows, startups, and universities and academic research groups.

AI Researcher | DICTA – The Israel Center for Text Analysis | 2022–present

I lead the center's NLP and LLM research on large collections of Hebrew and rabbinic texts: setting the research agenda, designing the methods, and taking systems from prototype to tools in active use by scholars. The work combines fine-tuned LLMs and custom text embeddings trained with contrastive learning for domain-specific corpora, powering information extraction, semantic search, text similarity, clustering, and question answering over long, complex documents, together with deep-learning models for morphological, syntactic, and semantic analysis of long documents at scale. I built the end-to-end training and inference pipeline, with tooling to analyze model outcomes and inspect the training process, and deployed the resulting document-understanding systems to production.

Example projects:

- Authorship attribution: built models that predict the author of texts with historically disputed attribution from textual signals alone, reaching high accuracy and resolving long-debated questions of authorship.
- Citation detection and cross-referencing: designed systems that automatically detect, extract, and link citations of classical sources across a large corpus, connecting documents that quote or paraphrase one another.

AI Researcher | Diagnostic Robotics | 2019

- Developed machine learning models for the healthcare domain, with applications in risk prediction (e.g., ER diagnoses, hospitalization forecasting).
- Collaborated with clinicians and researchers to refine AI-driven predictive healthcare decision-support systems.

AI Researcher | Amdocs | 2018–2019

- Led the development of a churn-prediction system for at-risk maintenance contract renewals, saving millions of dollars.
- Designed, built, and deployed the system from scratch, including problem definition, algorithm design, evaluation, and deployment pipeline.
- Presented the system and its business results to senior leadership at the company.

Research Assistant | Weizmann Institute of Science | 2011–2012

- Developed image-processing algorithms for 3D biomedical images.

Teaching & Advising

Ben-Gurion University of the Negev – Software and Information Systems Engineering Dept.

- Teaching Assistant, Databases – Fall–Spring 2016, 2017, 2018
- Teaching Assistant, Analysis and Design of Software Systems – Spring–Summer 2016, 2017, 2018
- Teaching Assistant, Intelligent Systems – Spring–Summer 2015
- Teaching Assistant, Decision Support Systems – Fall–Spring 2015
- Departmental Seminar Coordinator – 2015, 2016, 2017, 2018
- Final Projects Coordinator (4th Year) – 2015, 2016, 2017, 2018
- Final Projects Adviser (Undergraduate) – 2015, 2016

Tel-Aviv Yafo Academic College – Information Systems Dept.

- Lecturer, Databases – Spring–Summer 2015, Fall–Spring 2015
- Lecturer, Game Theory – Spring–Summer 2015
- Teaching Assistant, Databases – Spring–Summer 2014

Bar-Ilan University

- Teaching Assistant, Statistical Methods in Computer Science – Spring–Summer 2014
- Teaching Assistant, Object-Oriented Programming – Fall–Spring 2013, 2014
- Teaching Assistant, Machine Learning – Fall–Spring 2014
- Teaching Assistant, Introduction to Computing – Fall–Spring 2013

Shenkar College of Engineering and Design

- Teaching Assistant, Introduction to Programming Systems – Spring–Summer 2012
- Teaching Assistant, Introduction to Computer Science – Fall–Spring 2012
- Teaching Assistant, Introduction to Object-Oriented Programming – Fall–Spring 2012

College of Management (Academic)

- Teaching Assistant, Object-Oriented Programming – Spring–Summer 2012

High School Teaching (Israel Ministry of Education)

- Mathematics Teacher (2010–2013) – taught Bagrut levels 3–5 points, specializing in students with learning disabilities (via Nitzan Association).

Professional Service

- **Reviewer:** AAAI, IJCAI, AAMAS, KDD, CHI, EC, ECAI, Group Decision & Negotiation
- Developed voting algorithm for Best Paper selection (IAAI 2018)
- Organized departmental seminars at Ben-Gurion University

Publications

Journal Articles

- Gerhardt D., Marcowitz-Bitton M., Schuster W.M., Elmalech A., Suissa O., Mash M. (2026). Gendered Words and Patent Grant Rates: A Textual Analysis. Northwestern Journal of Technology and Intellectual Property, 23(3), Article 2.
- Raphael I, Xiong Z, Sneiderman CT, Raphael RA, Mash M., et al. The T cell receptor landscape of childhood brain tumors. Science Translational Medicine. 2025 Mar 19; 17(790): eadp0675.
- Mash, M., Fairstein, R., Bachrach, Y., Gal, K., & Zick, Y. (2020). Human-Computer Coalition Formation in Weighted Voting Games. ACM Transactions on Intelligent Systems and Technology (TIST).
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2017). Which Is the Fairest (Rent Division) of Them All? Journal of the ACM (JACM). *Selected as a Communications of the ACM Research Highlight, 2018.*
- Rochlin, I., Sarne, D., & Mash, M. (2014). Joint Search with Self-Interested Agents and the Failure of Cooperation Enhancers. Artificial Intelligence Journal (AIJ).

Conference Proceedings

- Mash, M., Oryol, S., Rosenthal, S., & Simmons, R. (2022). Predicting Data Scientist Stuckness During the Development of Machine Learning Classifiers. VL/HCC.
- Mash, M., Rosenthal, S., & Simmons, R. (2021). DSWorkFlow: A Framework for Capturing Data Scientists' Workflows. CHI.

- Mash, M., Bachrach, Y., Gal, K., & Zick, Y. (2017). How to Form Winning Coalitions in Mixed Human-Computer Settings. IJCAI.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2017). Which Is the Fairest (Rent Division) of Them All? IJCAI. (Best Sister Conferences track, extended abstract)
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2016). Which Is the Fairest (Rent Division) of Them All? EC. (Best Paper Award)
- Mash, M., Lin, R., & Sarne, D. (2014). Peer-Design Agents for Reliably Evaluating Distribution of Outcomes in Environments Involving People. AAMAS.
- Mash, M., Rochlin, I., & Sarne, D. (2012). Join Me with the Weakest Partner, Please. IAT.